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Entangled Up in Blue

Wave Functions of Many Parts

Popular discussions of the Einstein-Bohr debates often give the impression that Einstein couldn’t quite handle the uncertainty principle, and spent his time trying to invent clever ways to circumvent it. But what really bugged him about quantum mechanics was its apparent nonlocality—what happens at one point in space can seemingly have immediate consequences for experiments done very far away. It took him a while to codify his concerns into a well-formulated objection, and in doing so he helped illuminate one of the most profound features of the quantum world: the phenomenon of *entanglement*.

Entanglement arises because there is only one wave function for the entire universe, not separate wave functions for each piece of it. How do we know that? Why can’t we just have a wave function for every particle or field?

Consider an experiment in which we shoot two electrons at each other, moving with equal and opposite velocities. Because both have a negative electric charge, they will repel each other. Classically, if we were given the initial positions and velocities of the electrons, we could calculate precisely the directions into which each of them would scatter. Quantum-mechanically, all we can do is calculate the probability that they will each be observed on various paths after they interact with each other. The wave function of each particle spreads out in a roughly spherical pattern, until we ultimately observe it and pin down a definite direction in which it was moving.